Expt No.9 Dt:

ANALYSIS OF ENGINE TEST DATA USING NUMPY AND PANDAS

Aim: To write and execute a Python program for analyzing engine test data using NumPy and Pandas.

Interactive Development Environment (IDE) Used: Jupyter NoteBook
Programming Language Used : Python

Python Concepts used:

NumPy and Pandas are important Python libraries used for numerical computing and data analysis. NumPy provides functions for efficient numerical operations. Pandas provides data structures and functions for organizing, analyzing, selecting and processing tabular data. A Pandas DataFrame is a two-dimensional table consisting of rows and columns. In addition, Python supports indexing, conditional expressions and functions for processing data.

Python Concept	Example	Description
NumPy	np.mean()	Calculates the mean of numerical data.
Pandas	pd.DataFrame()	Creates a DataFrame from structured data.
DataFrame	df	Stores data in rows and columns.
Conditional Selection	np.where()	Selects values based on a condition.
Filtering	df[condition]	Filters rows based on a boolean condition.
List Conversion	to_list()	Converts a pandas Series to a standard Python list.

Data Analysis Operation / Concept:

Data Analysis Operation / Concept	Example Code	Description
Import NumPy	import numpy as np	Imports the NumPy library and assigns the alias np.
Import Pandas	import pandas as pd	Imports the Pandas library and assigns the alias pd.
Create DataFrame	pd.DataFrame(data)	Creates a DataFrame from structured data.
Calculate Mean	np.mean(df["Temperature_C"])	Calculates the arithmetic mean of a column.
Conditional Operation	np.where(condition, A, B)	Selects A when the condition is true and B otherwise.
Round a Value	round(value, 2)	Rounds numerical values to two decimal places.
Filter DataFrame	df[df["Status"] == "High Temperature"]	Selects rows where the status is "High Temperature".
Extract Column to List	df[...]["Engine"].to_list()	Extracts the Engine column from filtered data and converts to a list.

Python Program:

```
import numpy as np
import pandas as pd
# Engine test data
data = {
   "Engine": ["E1", "E2", "E3", "E4", "E5"],
   "Temperature_C": [85, 92, 105, 88, 110],
   "Fuel_Consumption_Lph": [2.5, 2.8, 3.2, 2.6, 3.5],
   "Power_kW": [5.0, 5.5, 6.0, 5.2, 6.5]
}
df = pd.DataFrame(data)
# Calculate overall averages using NumPy
avg_temp = np.mean(df["Temperature_C"])
avg_fuel = np.mean(df["Fuel_Consumption_Lph"])
avg_power = np.mean(df["Power_kW"])
# Identify engines exceeding the temperature limit
temperature_limit = 100
df["Status"] = np.where(
   df["Temperature_C"] > temperature_limit,
   "High Temperature",
   "Normal"
)
print("ENGINE PERFORMANCE ANALYSIS")
print(df.to_string(index=False))
print("\nAverage Temperature:", round(avg_temp, 2), "°C")
print("Average Fuel Consumption:", round(avg_fuel, 2), "L/h")
print("Average Power:", round(avg_power, 2), "kW")
print("\nEngines requiring temperature investigation:")
print(df[df["Status"] == "High Temperature"]["Engine"].to_list())
```

Sample Input:

Input: The engine test data are predefined in the Python dictionary in the program. No keyboard input is required during execution.

Engine	Temperature_C	Fuel_Consumption_Lph	Power_kW
E1	85	2.5	5.0
E2	92	2.8	5.5
E3	105	3.2	6.0
E4	88	2.6	5.2
E5	110	3.5	6.5

Expected Output:

Result: The Python program was written and executed successfully for analyzing engine test data using NumPy and Pandas, calculating average values, and identifying engines exceeding temperature limits.